

Recap

Web applications are distributed applications organized according to a client/server model

Client and server components communicate using the HTTP protocol

HTTP is an open stateless protocol based on request and response messages

The protocol has evolved over the years to cope with the technological evolution

All versions have many characteristics in common

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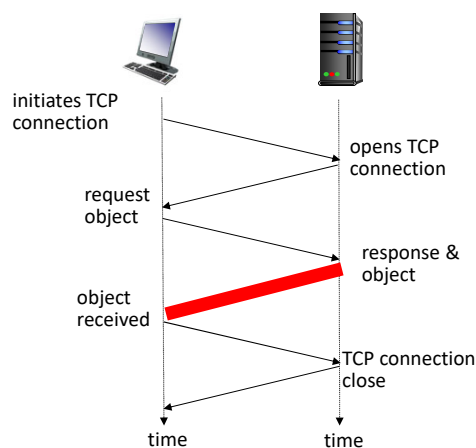
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Basic implementation

User opens a browser and specifies the URL of the object to be transferred

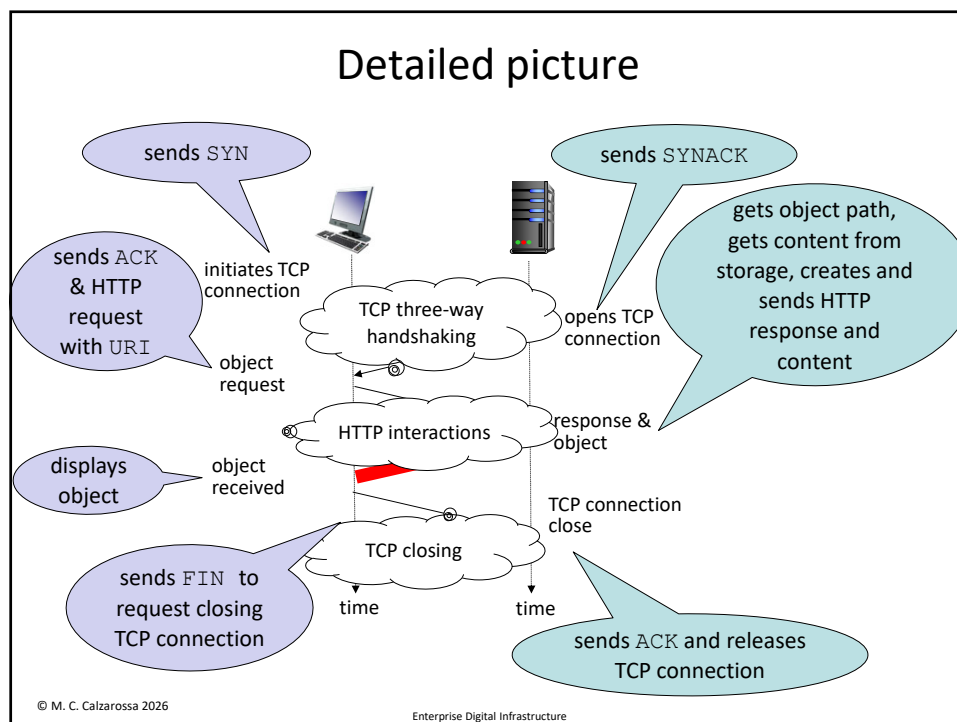
- Browser initiates TCP connection (creates a socket) to the server on port 80
- Server accepts TCP connection from client
- HTTP messages are exchanged between browser and web server
- TCP connection closed



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Page Load Time

Page Load Time (PLT) is the metric used to assess the efficiency of web applications and the quality of experience of the users

It includes the time for the HTTP interactions between client and web server and for the object transfer as well as the TCP overhead (and the TLS overhead)

The overheads refer to the time spent for the TCP and TLS handshakes

For “small” transfers these overheads might represent a significant fraction of the overall Page Load Time

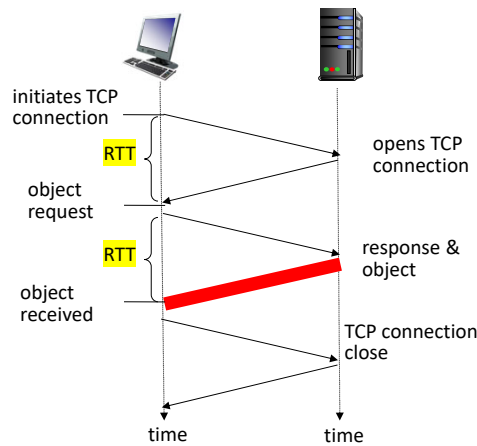
TCP has been designed for long lived connections

Web applications are elastic applications

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How long does it take to download a web object?



RTT: time for a “small” packet to travel from client to server and back (depends on propagation delay of the signal, queueing & processing delays at routers, number of hops, ...)

Transfer time: time spent to download (or upload) the object (depends on object size and available bandwidth)

Page Load Time =
RTT + RTT + transfer time

Closing time not part of PLT

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Google Drive home page: page source

```
<head>
<meta charset="utf-8" />
<title>Google Drive: Share Files Online with Secure Cloud Storage | Google Workspace</title>
<meta name="description" content="Learn about Google Drive's file sharing platform that prov
personal, secure cloud storage option to share content with other users." />
```

```
<link rel="stylesheet" href="/assets/SdHIXrh5.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/zPypI3Wr.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/BEwVz9nn.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/8nwFsvE3.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/DYv9Fyjj.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/CZx98z6f.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/Bctcio4i.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/DB6tyjNO.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/7WYUZQKH.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/Colj0pNL.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
```

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How long does it take to download a page?

The HTML file is transferred first (over a TCP connection) and then parsed by the browser to check the presence of referenced objects and extract their links

Before rendering the page, each object needs to be transferred (over a new TCP connection)

Page Load Time:

for the HTML file:

RTT + RTT + HTML transfer time + *time for HTML parsing*

for each referenced object:

RTT + RTT + object transfer time

All TCP connections should be closed

Significant operating system overhead for managing TCP connections

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How to improve PLT & efficiency

Reuse the TCP connection between a client and a server for multiple HTTP requests/responses (persistent TCP connection)

All objects transferred within a single TCP connection

Page Load Time:

for the HTML file:

RTT + RTT + HTML transfer time + *time for HTML parsing*

for each referenced object:

RTT + object transfer time

One RTT per object (after the HTML)

Benefits increase with the number of objects

Negligible operating system overhead for the TCP connection

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